

Getting Started with AWS CDK

Setup and Step-by-Step Instructions

Kickoff: AWS CDK Essentials



Define Cloud Infrastructure with Code

AWS CDK enables infrastructure as code using familiar languages.



Modern, Scalable Deployments

Automate and manage cloud resources for robust, repeatable setups.



Getting Started: Setup & First Steps

Begin with prerequisites, CLI installation, and project initialization.

Prerequisites for Success

Install Node.js 18+ and npm or yarn

Ensure your system has the latest Node.js runtime.

Set Up Your AWS Account

Sign up for AWS; Free Tier is sufficient for learning.

Install and Configure AWS CLI

Download AWS CLI, configure credentials, and verify setup.

Basic JavaScript/Node.js Knowledge

Familiarity with JavaScript helps in writing CDK code.

Setting Up AWS CLI



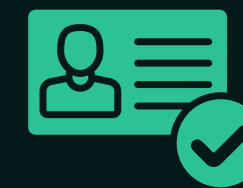
Install AWS CLI

- Download and install from AWS official site
- Verify installation with `aws --version`



Configure AWS Credentials

- Run `aws configure` to set up access keys
- Input AWS Access Key, Secret Key, region, and output format



Test AWS CLI Access

- Use `aws sts get-caller-identity` to confirm setup

Installing AWS CDK CLI

Install AWS CDK CLI Globally

Run: `npm install -g aws-cdk`

Verify Installation

Check version: `cdk --version`

Create Your First CDK Project



Set Up Your CDK Workspace

- Create a new directory for your project
- Navigate into the directory



Initialize a TypeScript CDK App

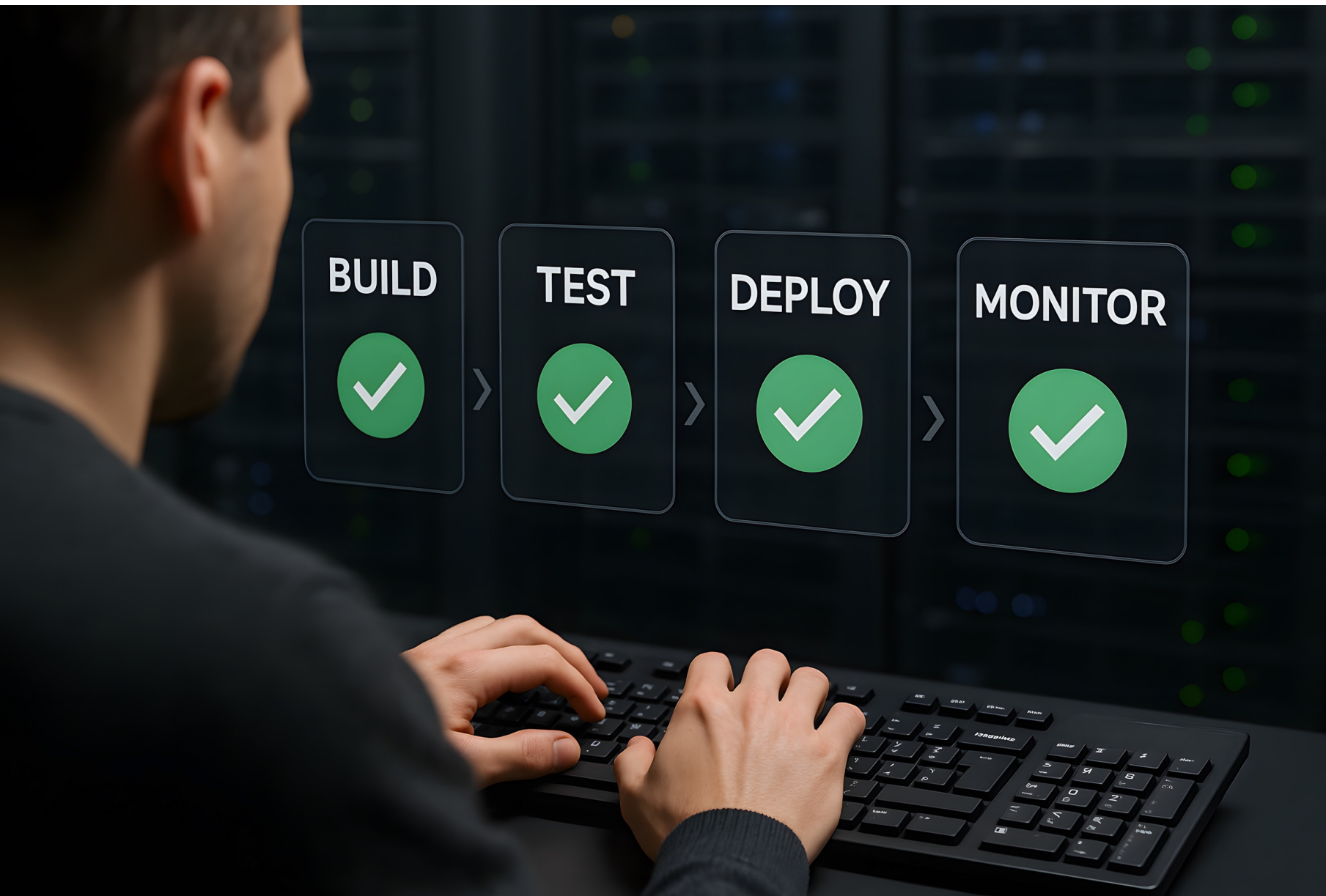
- Run: `cdk init app --language=typescript`



Explore the Project Structure

- `bin/`: Entry point for the CDK app
- `lib/`: Define infrastructure stacks
- `cdk.json` & `package.json`: Configuration and dependencies

Bootstrapping Your AWS Account



Run 'cdk bootstrap' Once Per Account/Region

- Prepares your AWS environment for CDK deployments

Sets Up Deployment Resources

- Creates an S3 bucket for storing deployment assets
- Establishes IAM roles for secure resource management
- Deploys the CDK toolkit stack for future operations

Enables Secure, Repeatable Deployments

- Ensures your account is ready for production-grade infrastructure
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Validating Your Setup

List Stacks

Run 'cdk ls' to view available stacks.

Synthesize CloudFormation

Use 'cdk synth' to generate CloudFormation templates.

Deploy Stack

Execute 'cdk deploy' to launch resources in AWS.

Key Takeaways & Next Steps



CDK Translates Code to CloudFormation

Infrastructure is defined in code and deployed as CloudFormation stacks.

Setup & Bootstrapping Are Critical

Proper setup ensures secure, repeatable deployments in AWS.

Standard Project Structure

Organized folders and files support best practices and scalability.

Ready for Serverless Production

Environment is prepared for robust, production-grade deployments.
